

Nasser Al-Ali

Statistical Programmer & Biostatistician | AI/Agentic Systems for Clinical Trials

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Summary

Statistical programmer and emerging biostatistician (PhD candidate, Statistics) with 6+ years in biotech/pharma clinical development — ADaM datasets, Tables/Figures/Listings (TFLs), CDISC SDTM/ADaM, and independent double-programming QC across oncology and other therapeutic areas. Over the past year, designed and built agentic AI systems that author and independently QC SDTM/ADaM deliverables and draft statistical analysis plans under deterministic, auditable gates — pairing deep clinical-statistics domain knowledge with modern LLM/agent engineering. Advanced SAS and R, Python; partners across Biostatistics, Data Management, Safety, and Regulatory under GCP/ICH.

Technical Skills

Clinical Programming: SDTM/ADaM datasets, TFLs, SAP-driven analysis, independent double-programming QC, CRF vs non-CRF reconciliation, Pinnacle 21 conformance, traceability

Languages: SAS (Advanced — Macro, SQL), R (Advanced — reusable functions, R-Shiny), Python, SQL

AI / Agentic Engineering: LangGraph, LangChain, Retrieval-Augmented Generation (RAG), ChromaDB / vector retrieval, Claude Agent SDK, multi-agent orchestration, LLM evaluation & guardrails, human-in-the-loop pipelines, FastAPI, DuckDB

Statistical Methods: Bayesian modeling, mixture models / IRT, survival analysis (OS, PFS; ADTTE), statistical & machine learning

Standards & Tools: GCP, ICH (incl. E9(R1) estimands), TIBCO Spotfire, Git/GitHub, Docker, GCP

Professional Experience

Denali Therapeutics, Senior Clinical Programmer (Contract)

South San Francisco, CA
Sept 2023 – July 2024

- Partnered with Biostatistics and Statistical Programming on ADaM-aligned specifications, TFL shells, and analysis outputs; performed independent QC and drove discrepancy resolution.
- Developed and validated R and SAS programs for data-quality checks, interim/exploratory outputs, and ad-hoc analyses under biostatistician direction.
- Built reusable R functions and SAS macros that standardized recurring study-reporting workflows and reduced rework across cross-functional teams.
- Reviewed CRO and internal biometrics deliverables for completeness, consistency, and traceability across Biostatistics, Clinical Operations, Data Management, Safety, and Regulatory.

Genentech (Roche Group), Clinical Data Analyst

South San Francisco, CA
Sept 2015 – July 2019

- Member of a late-phase oncology clinical programming group; created and validated SDTM and ADaM datasets and TFLs supporting oncology trials using SAS and R.
- Reconciled CRF and non-CRF data to ensure data integrity and regulatory readiness across complex multi-source datasets.
- Built interactive R-Shiny and TIBCO Spotfire dashboards to communicate study metrics to clinical and executive stakeholders.
- Supervised interns and FSP personnel, reviewing deliverables and enforcing data-quality standards.

University of Nevada, Reno, Research & Teaching Assistant (Statistics)

Reno, NV
Aug 2020 – June 2023

- Conducted NSF-funded statistical research (Bayesian modeling, computational statistics, time-series) advised by Dr. Ilya Zaliapin; taught undergraduate statistics courses.

XOMA, Statistical Programming Analyst (Intern)

Berkeley, CA

- Validated and generated ADaM analysis datasets per specifications and SAPs; produced TFLs using double-programming methodology.
- Supported biostatisticians with SAP creation and ad-hoc/post-hoc analyses across multiple therapeutic areas.

Feb 2014 – May 2014

Selected Projects (AI for Clinical Biometrics)

Agentic Biometrics Department — raw to SDTM to ADaM

- Designed a "virtual biometrics department" of scoped AI subagents that author specs and production SAS and independently double-program CDISC datasets — every deliverable gated by PROC COMPARE, Pinnacle 21 conformance, and SAS-log checks before a named human signs off ("validate the wrappers, not the brain").
- On a synthetic oncology study, drove the Demographics domain end-to-end (raw to SDTM, independent double-programming to a clean PROC COMPARE, SYSINFO=0) and scaffolded 20 SDTM + 6 ADaM work orders; plugin tooling covered by 107/107 automated tests.

Protocol-to-SAP Generator

- Built a multi-agent (LangGraph) system that drafts a sponsor-grade Statistical Analysis Plan from a protocol PDF with ICH E9(R1) estimands and a traceability appendix, grounded via RAG in FDA/EMA/ICH guidance; internal evaluation across 8 public protocols scored 83.75/100 (96/98 tests passing).

Study-Data Chatbot

- Built a self-hosted assistant that answers plain-English questions over a study's SDTM/ADaM data via read-only, audited SQL (the model never computes a number), with DRAFT watermarking and a one-click hand-off into the double-programming QC pipeline; 33/33 backend tests passing.

Education

PhD University of Nevada, Reno, Statistics & Data Science

Reno, NV

- Can-** • Dissertation: Bayesian Zero-Inflated Unipolar Mixture Item Response Theory Model with Covariate-Dependent Latent Classification — Bayesian computation and simulation, with applications to patient-reported outcomes. Expected August 2026.
- didate**

Aug 2020 – Aug 2026

MS California State University, East Bay, Statistics

Hayward, CA

June 2016

BS Khalifa University, Computer Engineering

Abu Dhabi, UAE

Jan 2014

Publications & Presentations

A Bayesian Zero-Inflated Mixture Item Response Theory Model for Patient-Reported Outcomes

2026

Al-Ali, N.

(WNAR (International Biometric Society) Annual Meeting, Pullman, WA)

Covariate-Dependent Latent Classes in Unipolar Mixture Item Response Models: A Bayesian Approach

2026

Al-Ali, N.

(International Meeting of the Psychometric Society (IMPS), South Korea)

SDTM Bookmarking Automation Leveraging SAS, Ghostscript, and Form-Visit Study Data

Sept 2016

Al-Ali, N., Paz, K.

(Western Users of SAS Software (WUSS) Conference Proceedings)